

Protocol Design Document

1. Overview

The quiz system uses a custom text-based application protocol over TCP sockets. Each message is a string with a clear type, but questions and leaderboard are displayed in block format for better readability.

2. Message Types

Client → *Server*

1. GENRE Request
GENRE|<genre_name>
2. Answer Submission
ANSR|<option_letter>
3. Leaderboard Request
LEAD_REQ
4. Keep-alive Response
ALIVE_OK
5. Exit Request
exit

Server → *Client*

1. Question Delivery (block format)
--- QUESTION ---
<question_text>
A) <option_A>
B) <option_B>
C) <option_C>
D) <option_D>

2. Answer Feedback
> [CLIENT] Correct
or
> [CLIENT] Wrong
3. Leaderboard Response (block format)
--- LEADERBOARD ---
<Client1>:<score>
<Client2>:<score>
<Client3>:<score>

4. Keep-alive Ping
KEEP_ALIVE
5. Informational Message
INFO|<message_text>
6. Error Message
ERROR|<error_text>

3. Protocol Differences: 1(A) vs 1(B)

- 1(A) (Without Cache):
Every GENRE request triggers new quiz generation.
- 1(B) (With Cache):

Server checks cache before generating.
- Cache hit → INFO|Cached questions used for genre <name>
- Cache miss → INFO|Fresh questions generated for genre <name>

4. Example Client–Server Exchanges

Session Start (1A)

```
CLIENT → SERVER: GENRE|science
SERVER → CLIENT:
--- QUESTION ---
Which planet is closest to the Sun?
A) Earth
B) Mercury
C) Venus
D) Mars
-----
CLIENT → SERVER: ANSR|B
SERVER → CLIENT: > [CLIENT] Correct
```

Leaderboard Request

```
CLIENT → SERVER: LEAD_REQ
SERVER → CLIENT:
--- LEADERBOARD ---
Client4:1
Client5:0
Client6:0
-----
```

Keep-Alive

```
SERVER → CLIENT: KEEP_ALIVE
CLIENT → SERVER: ALIVE_OK
```

Cache-Aware Exchange (1B)

```
CLIENT → SERVER: GENRE|history
SERVER → CLIENT: INFO|Cached questions used for genre history
SERVER → CLIENT:
--- QUESTION ---
Who was the first US president?
A) Lincoln
B) Washington
C) Jefferson
D) Adams
-----
CLIENT → SERVER: ANSR|B
SERVER → CLIENT: > [CLIENT] Correct
```

Exit

```
CLIENT → SERVER: exit
SERVER → CLIENT: INFO|Quiz over - final scores available
<connection closed>
```